MARLF: Multi-Agent Reinforcement Learning Framework for Distributed Autonomous Decision Making in Dynamic Environments

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ABSTRACT

The rapid growth of intelligent autonomous systems has increased the demand for efficient decision making mechanisms in complex and dynamic environments. Traditional centralized decision frameworks often face limitations in scalability, adaptability, and communication overhead when deployed in distributed systems. Multi agent reinforcement learning has emerged as a promising approach for enabling autonomous agents to learn cooperative strategies through interaction with the environment. This paper investigates a distributed multi agent reinforcement learning framework designed to improve collaborative decision making among autonomous agents operating in dynamic conditions. The proposed framework focuses on efficient policy learning, adaptive coordination, and improved system scalability. A comprehensive literature survey is conducted to analyze existing techniques, identify research gaps, and highlight the challenges associated with distributed autonomous decision making.

1. Introduction

The rapid advancement of intelligent computing systems has significantly increased the demand for autonomous decision making in dynamic and distributed environments. Modern applications such as autonomous vehicles, underwater robotic systems, and large scale Internet of Things infrastructures require intelligent agents capable of analyzing environmental information and making adaptive decisions in real time. Traditional centralized control approaches often face limitations in scalability, communication overhead, and delayed response when deployed in complex distributed systems. As a result, researchers have increasingly focused on decentralized and learning based frameworks that allow autonomous agents to operate collaboratively while adapting to continuously changing environmental conditions [32, 3].

Recent survey studies have highlighted the growing importance of intelligent coordination and distributed learning techniques in modern autonomous systems. For instance, the survey presented by Wu et al. analyzes cooperative path planning approaches for autonomous underwater vehicle clusters and discusses the evolution of heuristic optimization, deep reinforcement learning, and graph based coordination mechanisms. These approaches enable multiple agents to collaboratively explore environments and optimize navigation strategies while maintaining communication efficiency and operational scalability in complex mission scenarios [32].

Similarly, the development of autonomous vehicle technologies has introduced new challenges related to safety validation and system evaluation in dynamic environments. Chen et al. provide a comprehensive review of scenario based virtual testing frameworks designed for automated vehicles. Their study discusses different simulation architectures, hazard evaluation mechanisms, and scenario generation techniques used to validate intelligent driving systems under diverse environmental conditions. Such testing frameworks play a crucial role in assessing the reliability and decision making capabilities of autonomous systems before real world deployment [3].

In addition to intelligent agents and learning algorithms, distributed computing infrastructures are essential for supporting large scale autonomous systems. Fog computing has emerged as an important paradigm that extends cloud capabilities toward the network edge, enabling low latency data processing and real time system responsiveness. A systematic literature review conducted by Aldossary et al. examines various load balancing strategies in fog computing architectures and highlights the importance of efficient resource allocation for supporting data intensive and latency sensitive applications such as Internet of Things based intelligent systems [2].

Although significant progress has been achieved in autonomous systems, distributed computing, and intelligent decision making frameworks, several challenges remain in designing scalable and adaptive solutions for dynamic environments. Existing survey studies emphasize issues related to coordination complexity, communication overhead, system scalability, and environmental uncertainty. These limitations highlight the need for improved frameworks capable of enabling collaborative decision making among multiple intelligent agents operating in distributed environments. Motivated by these challenges, this study investigates a multi agent reinforcement learning framework for distributed autonomous decision making in dynamic environments.

The rapid expansion of cyber physical systems and industrial internet based infrastructures has significantly transformed modern intelligent environments. As industrial systems become increasingly interconnected, the need for reliable monitoring and intelligent security mechanisms has

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become more critical. Hosseini et al. present a systematic review of artificial immune system based approaches for industrial intrusion detection and cyber defense frameworks. Their study highlights how biologically inspired learning models can detect abnormal behaviors and enhance resilience in complex industrial networks. The survey also discusses several detection strategies and evaluation mechanisms that support autonomous monitoring and adaptive security responses in highly dynamic computing environments [13].

Similarly, the evolution of distributed computing paradigms has introduced new architectural models that support real time processing at the network edge. Fog computing has emerged as a promising solution that extends cloud services closer to data sources, enabling low latency processing and efficient resource utilization for internet of things applications. Aldossary et al. provide a systematic literature review on load balancing strategies in fog computing environments and analyze different architectures designed to improve workload distribution. Their study emphasizes that efficient task allocation mechanisms are essential for maintaining system stability, improving processing efficiency, and supporting intelligent distributed decision making in heterogeneous computing environments [2].

Furthermore, the increasing complexity of distributed intelligent systems requires advanced coordination strategies that enable autonomous decision making across multiple interconnected components. Recent research highlights the importance of adaptive learning models and distributed control mechanisms in managing large scale environments where centralized control becomes inefficient. Studies on distributed computing and intelligent system architectures emphasize the integration of learning based techniques with decentralized infrastructures to improve system scalability and operational efficiency. These developments provide a strong foundation for designing multi agent learning frameworks capable of supporting cooperative decision making and adaptive behavior in complex dynamic environments [13, 2].

1.1. Research Objectives

The goal of this study is to build a distributed multi-agent reinforcement learning system that lets agents make their own decisions in changing and unpredictable situations, without needing someone else to watch over them. Unlike systems that follow fixed rules or rely on a central controller, this method uses decentralized learning. Each agent learns from what it sees and interacts with directly. This setup helps the system work better when dealing with complex situations where a central control isn't possible or practical.

The study also wants to add smart learning features to the MARL system so it can handle changes in multi-agent environments. When many agents are learning and improving their strategies at the same time, the environment each agent experiences keeps changing. To address this, the research focuses on making agents able to adjust their learning methods on the fly. This helps them learn more steadily, reach good solutions faster, and make better decisions over time in changing environments.

The study also wants to create good ways for agents to communicate and work together with low computational needs. Even when they don't have full information about each other or the environment, the system aims to improve how well they can make group decisions without using all dataportant for real-world systems where sending and receiving data is limited and scaling up is a challenge.

Finally, the study will test the new system in realistic and changing simulation environments. The system's success will be measured by how well it learns, how quickly it adapts to changes, how well it handles more agents, and how effectively it makes decisions. These tests will show how well the method works for real-world autonomous systems.

2. Literature Survey

Heuristic optimisation [32], deep reinforcement learning, and graph neural network-based distributed control are the three main paradigms that have shaped research on AUV cluster cooperative path planning hasAUV class cooperative path planning over the last 20 years. In addition to addressing particular facets of the path planning issue, each paradigm also highlights particular drawbacks.Heuristic algorithms are very accurate in static optimisation tasks, but they are not flexible enough to work in dynamic marine environments. Although reinforcement learning improves decision autonomy and environmental responsiveness, it is still constrained by data efficiency and transferability to real-world scenarios. Scalability and communication efficiency are enhanced by distributed control frameworks and graph neural networks.

A significant advancement [29] in the integration of autonomous robotic systems and digital twins is the Distributed Multi-Agent Digital Twin Environment suggested in this study. Its framework's adaptability and scalability enable effective management of individual agents or large robot fleets with real-time synchronisation, performance optimisation, and risk minimisation in dynamic environments.The ability of DMDTE to model and regulate higher automation systems in a way that allows operating strategies to be optimised with efficiency and non-negligible responsiveness is its greatest strength. DMDTE is special because it can adapt in real time to any operational environment by combining real-time simulation with physical agent interaction.

An autonomous task allocation problem for large-scale, dynamic spatiotemporal sensing by a swarm of autonomous drones is examined in this paper[26].In order to optimise flight paths, data collection, and recharging assignments, a novel hybrid optimisation technique called DO-RL is put forth that combines distributed optimisation and DRL. The efficacy of DO-RL is demonstrated by extensive experimentation, which demonstrates how combining short-term and long-term strategies enhances sensing performance, energy

management, and environmental adaptability. Extended operations and effective resource management are made possible by the suggested method. Nevertheless, there are still certain drawbacks, such as the dependence on centralised training, security issues with decentralised coordination, and the requirement that future studies take multi-modal sensing and environmental limitations into account.

In this paper[19, 30] , we have presented a comprehensive survey of the applications of MARL in the future internet. First, we have introduced the backgrounds, challenges, and advanced algorithms of single-agent reinforcement learning and multi- agent reinforcement learning. Afterward, we have reviewed, analyzed, and compared the recent research contributions that use MARL to address the emerging issues in the future internet. These issues involve network access, transmit power control, computation offloading, content caching, packet rout- ing, trajectory design for UAV-aided networks, and network security. Finally, we have outlined the key challenges, open issues, and future directions that can help the readers to further explore research topics in this area.The proposed PCA-MDP approach performs better than the conventional load-balancing approaches since it offers improved load harmony, adaptiveness, and efficiency of the resources at hand. The proposed approach takes into account the varying load states of nodes and distributes the resources evenly so as to avoid the problem of overload and subsequently increasing the stability of the entire system. This adaptability is mainly beneficial to real time applications like CAVs, Augmented Reality, and IoT based networks. The PCA-MDP model has proven to be successful in the farther improvement of load harmony and reducing load variance across edge computing nodes. With the combination of probabilistic cellular automata with the Markov decision process the model enables a more optimal arrangement of the tasks, improving performance of the system as a whole.

This paper [33] presents a distributed task allocation method for multi-UAV persistent monitoring in dynamic environments. The method features a GMM-based reward map that aggregates spatial-temporal reward, an energy-aware mechanism that integrates charging requirements into the reward structure, and a communication-driven task coordination approach that maintains system resilience under limited communication. The proposed method enables efficient task allocation while handling energy constraints and communication limitations without explicit scheduling or recovery protocols. Experimental results demonstrate the effectiveness of our approach, showing superior scalability with consistent performance as system scale increases. The communication-driven coordination demonstrates robust adaptability to both communication constraints and UAV failures.

This paper [27] presented and experimentally tested the ARCog-NET architecture, a distributed cognitive architecture that is intended to provide support to the creation of intelligent and scalable multi-robot swarms. The framework was designed into a generic and adaptable structure in that it could be implemented in different fields, such as mapping, search and rescue and environmental monitoring. ARCog-NET was implemented with Monocular Visual SLAM and performed in a real-life experiment of mapping an indoor environment via a swarm of six UAVs. The experiment has been shown to exhibit more effective and powerful swarm behavior in the form of decision weight learning, historical knowledge reuse and layered coordination.

This research [34, 25] presented a novel cyber-physical systems framework for autonomous mobile robot integration in warehouse environments, demonstrating transformative potential for intelligent inventory management. The proposed hierarchical architecture successfully addressed critical challenges in multi-robot coordination, real-time optimization, and human-robot collaboration through digital twin synchronization and distributed decision-making mechanisms. Experimental validation confirmed substantial improvements in operational efficiency, with 420 orders per hour throughput, 99.7 inventory accuracy, and 34 reduction in energy consumption compared to state-of-the-art alternatives. The economic viability demonstrated through 11–18 month payback periods and scalability to 30+ robot deployments positions this technology for widespread adoption across the logistics industry.In the proposed network model, two elements make it possible to manage the increase in the average travel time generated as the number of drones on the network increases, while ensuring safe conditions: dynamic links and headway time. As can be seen from Table 5, the average travel time between two fixed nodes tends to increase as the number of drones increases. The presence of disturbances represents a more realistic operating condition and further increases the travel time. To conclude, the 3D-UAN network makes it possible to manage an increasing number of drones effectively, thanks to the defined architecture that introduces distributed rules, vertical stratification, and strategic planning together with autonomous and distributed operational control. By combining a 3D-UAN with a scalable agent-based control system, this research provides a fundamental framework for exploring complex urban logistics scenarios.

The convergence of GenAI and IoT [1] represents a transformative shift in the development of intelligent, adaptive, and autonomous systems across sectors such as healthcare, smart cities, transportation, and industrial automation. This integration holds immense promise in enhancing data processing, real-time decision-making, personalization, and system resilience. By leveraging capabilities such as synthetic data generation, federated learning, and edge intelligence, GenAI enables more efficient and context-aware IoT applications. However, realizing the full potential of this synergy entails addressing challenges spanning computational constraints, data privacy, interoperability, energy efficiency, and ethical concerns. Ethical imperatives necessitate the adoption of privacy-preserving frameworks and explainable AI to ensure trust and accountability.

This study [23, 12] presents a context-aware trust evaluation framework for IoT nodes operating in dynamic and

mission-critical environments. Leveraging supervised learning, specifically Decision Tree Regression, the model estimates node reputation scores based on behavioral metrics including transmission success rate, packet loss, latency, battery level, and peer feedback. A physiological adjustment mechanism using a Medical Stress Index was introduced to integrate vital signs and stress level into a dynamic penalty system. The adjusted model preserved predictive accuracy while introducing greater resilience and fairness. The framework performs consistently on new data and offers a scalable and responsible solution for secure, real-time decision-making in battlefield IoT environments.This research introduces a novel methodology that combines fuzzy inference systems with real-valued genetic algorithms to optimize decision-making in a multi-agent control environment. The proposed approach is applied to the RoboCup Soccer Simulation 2D League, demonstrating how interpretable AI and evolutionary computation can be integrated in real-time strategic planning. A key contribution is the design of an evaluation function that enables autonomous agents to adaptively assess their positional advantage using fuzzy rules learned through evolutionary processes. The interpretability of the learned behavior allows visualization and understanding of agent decisions. Collectively, these contributions provide a practical, interpretable, and scalable approach to enhancing decision-making in cooperative, uncertain, and dynamic multi-agent systems.

Aiming to solve the problem of the UAV [9, 20] command and control system's inability to perform intelligent decision-making and resource allocation, this paper proposes a "cloud, edge, and end" command and control system based on a large language model. The system enhances intelligent decision-making, environmental adaptation, and resource allocation abilities through a three-layer distributed architecture empowered by large language models. Large-scale simulation experiments and practical scenario verification demonstrate the effectiveness of the system, with task latency, energy efficiency, and resource utilization improved significantly. Despite these advantages, challenges remain in computational resource requirements, communication bandwidth dependence, and system standardization. With the continuous maturation of related technologies, this system is expected to play an important role in more fields.This paper focuses on the problem of distributed autonomous decision-making for fixed-wing UAVs in pursuit–evasion games based on multi-agent reinforcement learning. A Markov PEG model is established, and systematic training and validation are conducted using the MAPPO algorithm within the CTDE framework. Simulation results demonstrate that the proposed reward structure effectively guides the pursuers to learn efficient strategies, achieving a high capture success rate. The introduced subgame curriculum learning mechanism facilitates the training process by progressively guiding the agents from simple to more complex tasks. Testing and analysis confirm the control effectiveness of the trained model in practical pursuit–evasion missions, highlighting the feasibility and potential of the proposed method.

Constructing architectures for [21] real-time ship control and navigation computing systems is paramount to safety, efficiency, and autonomous control over ships. Hybrid computing offers the most balanced functionality by combining the responsiveness of edge systems and the cloud's computing strength. Edge computing provides low latency, high reliability, and complete functionality during network loss, while cloud computing is powerful in analytics and control. Cloud-augmented edge computing achieves high availability, accurate prediction and data fusion, and robust uptime during network outages. Such capabilities are extremely important in dynamic maritime scenarios, making the hybrid approach ideal for scalable autonomous ship navigation and control systems.

The integration of sensor [31] fusion techniques and reinforcement learning-based FTC algorithms enables AMVs to proactively adapt to faults, thereby reducing operational downtime and enhancing mission reliability. Incorporating the Internet of Robotic Things facilitates distributed fault monitoring, cooperative decision-making among AMV fleets, and secure cloud-based diagnostics, ensuring efficient long-term operations. Simulation-based validation demonstrated that the learning-based approach significantly improves fault detection accuracy, system adaptability, and recovery efficiency compared to traditional rule-based FTC methods. The proposed framework paves the way for self-learning, resilient AMVs capable of handling unforeseen system failures without human intervention and supports collaborative operation using IoRT and ROS-enabled robotic systems.

Mobile mobility management [4, 15] represents a novel shift in understanding user interaction under varying mobile network conditions but faces major implementation challenges. Privacy and security issues arise due to sensitive contextual information such as user location, identity, and usage, necessitating strong anonymization, encryption, and adaptive access control mechanisms. Scalability is another critical concern as the exponential growth of IoT devices and mobile users overwhelms traditional and centralized cloud-based architectures. This highlights the need for scalable and responsive frameworks based on edge computing and distributed processing. Additionally, seamless integration with legacy systems and cross-vendor compatibility remains challenging, requiring middleware solutions and standardized approaches to fully realize context-aware mobility management.In this study, we proposed the V2X-Car Edge Cloud, a cloud-native platform designed to address challenges in real-time processing, scalability, and security within latency-sensitive V2X applications. By integrating DevSecOps automation principles, cloud-native Kubernetes orchestration, and hybrid communication protocols, the system demonstrated secure deployment, dynamic resource allocation, and real-time monitoring capabilities across decentralized edge environments. Validation revealed improvements in latency, resource management, and operational stability using virtual

driving simulations. The proposed platform enables cost-effective validation of autonomous driving services, improves network transparency, and supports real-time object detection and data analysis. However, limitations include hardware dependency and lack of testing under extreme conditions, indicating the need for further real-world validation and scalability, latency, and security benchmarking in future work.

In this study [21], we tackled the challenge of optimizing strategies in dynamic and complex multi-agent systems, where traditional game-theoretic methods fall short due to their static assumptions and inability to adapt to real-time interactions. We proposed a hybrid deep reinforcement learning framework combining two main components including the Strategic Interaction Neural Encoder and the Adaptive Strategic Equilibrium Optimization algorithm. Experimental results demonstrated that this framework enhances adaptability in evolving environments and improves convergence efficiency and strategic robustness compared to baseline models. However, limitations remain related to sensitivity to hyperparameter configurations, training data quality, and the assumption of complete state observability, highlighting the need for uncertainty modeling and transfer learning in future work.

This paper [5] proposes a distributed architecture algorithm based on MARL to adaptively adjust the CW in a network environment with nodes operating at different transmission rates. The experimental results show that the proposed distributed architecture algorithm outperforms both the centralized architecture method and the traditional BEB algorithm in terms of overall performance. In static scenarios, the distributed algorithm yields up to 30% higher throughput when the network size ranges from 10 to 40 nodes, and remains consistently higher as the number of nodes increases. This improvement is mainly due to the algorithm's ability to adaptively adjust CW values based on varying transmission rates. Dynamic network experiments demonstrate that the distributed architecture can quickly adjust transmission strategies without starvation behavior, although fairness and higher training cost remain issues.

To address the issues [17]of low efficiency and poor coordination in multi-UAV transport tasks, we propose an IMAHAC algorithm developed within a virtual logistics environment based on the OpenAI Gym system. The IMAHAC algorithm integrates a hybrid attention mechanism with the TT-PER prioritized replay mechanism and embeds them into the MAAC framework. Experimental results demonstrate that IMAHAC outperforms baseline algorithms and proves effective in multi-UAV logistics scenarios. The sensor fusion layer in the actor network enables efficient utilization of diverse sensor data, while the dissimilarity layer in the critic network improves collaboration using cosine similarity. Under identical conditions, IMAHAC transported more cargo than MAAC, demonstrating superior collaboration capabilities and enhanced obstacle avoidance performance.

This paper [3, 8] provides a comprehensive review of the key technologies and methodological frameworks for the virtual testing of automated wheeled vehicles and the challenges in transferring these technologies from road to non-road environments. Examining scenario risk evaluation, hazardous scenario generation, and accelerated evaluation revealed that while a mature evaluation framework exists for road-based autonomous vehicles, significant challenges arise for non-road vehicles, including lack of unified risk evaluation criteria and insufficient foundational data. To address these issues, this paper proposes a testing adaptation approach driven by multi-physics simulation and digital twin technologies, forming a holistic framework for virtual testing. Future work should focus on algorithm development tailored to specific scenarios to advance the transfer framework toward practical validation.it has presented a fully integrated, AI-driven, latency-aware framework for decentralized decision-making in IoT-connected heterogeneous Multi-Robot Systems, combining deep learning perception, 5G-enabled communication, and energy-optimized task execution. The proposed Faster-RCNN-Pose architecture enables simultaneous object detection and 6-DoF pose estimation, reducing perception-to-decision latency while maintaining high accuracy in dynamic environments. Real-time evaluations confirmed robust detection, reliable pose estimation, and operational effectiveness under bounded 5G latency. The system incorporates latency-aware task allocation, hybrid centralized–decentralized control, and energy-aware decision-making, improving synchronization, mission continuity, and resource efficiency. This unified architecture advances scalable, resilient multi-robot coordination for real-world autonomous applications.

This survey [16, 7]provided a comprehensive review of the ISCC paradigm for the IoV. The developmental trajectory of the IoV demonstrates that overcoming the limitations of traditional separate systems requires moving beyond pairwise integration and toward a holistic ISCC approach. To this end, this survey charted the path to this holistic paradigm by classifying the evolution of ISCC in the IoV into distinct levels of integration and reviewing key enabling technologies. Existing ISCC research in IoV was analyzed and classified into task-oriented and system-level approaches. The discussion of remaining open challenges and proposed solutions offered a clear roadmap to guide future research endeavors in this field. we explored the application of cooperative MARL within to address efficient data gathering under diverse constraints and dynamic conditions. The successful implementation of MARL signifies a substantial leap forward compared to traditional data-gathering methodologies. Sensors function as independent yet cooperative agents within a decentralized framework, enabling adaptive optimization of data transmission, particularly under energy-harvesting constraints. The study concludes that stochastic policies are essential for handling non-deterministic constraints and uncertainties in complex WSN scenarios. Simulation and software-defined radio implementations validated the effectiveness of the proposed MARL model, demonstrating high channel utilization and adaptability under stringent resource constraints.

Table 1
Literature Survey Summary

Ref.	Method Proposed	Techniques Used	Issues Resolved	Limitation
[18]	Bidirectional integration of Generative AI and mobile networking	Channel modeling, transmission optimization, network slicing, mobility-aware handover, generative AI	Improved intelligence and adaptability of mobile network management; support for user-centric services such as extended reality and autonomous mobility	Future mobile network architectures and data management must evolve further to fully support GAI
[32]	Cooperative path planning for AUV clusters	Heuristic optimization, deep reinforcement learning, graph neural networks, distributed control	Improved adaptability, decision autonomy, communication efficiency, and scalability in AUV cooperative path planning	Limited adaptability in dynamic environments, data inefficiency in reinforcement learning, and insufficient robustness at large scale
[14]	Integration of Large Language Models with 3D vision for robotics	Spatial grounding architectures, multimodal sensor fusion, open-vocabulary pre-training, text-to-3D generation	Enhanced object referencing, localization, situational awareness, and generalization under data scarcity	Challenges in real-time adaptation, modality gaps between 2D and 3D data, and evaluation limitations
[24]	Load-balancing strategies in fog computing	Three-layer IoT–Fog–Cloud architectures, heuristic and meta-heuristic algorithms, machine learning, reinforcement learning	Reduced latency, improved energy efficiency, better resource utilization under dynamic workloads	Lack of real-world deployments limits generalizability and practical applicability
[2]	Future-oriented fog computing load-balancing frameworks	Energy-aware computing, context-aware scheduling, security integration, multi-objective optimization, benchmarking frameworks	Improved sustainability, adaptability to mobility and QoS demands, enhanced security and privacy	Overreliance on simulations and absence of standardized evaluation frameworks
[22]	Unified systems framework for intelligent and resource-efficient systems	Control theory, distributed self-organizing systems, hierarchical decision-making, multiobjective optimization	Efficient use of scarce resources, improved scalability, learning, and autonomous decision-making	High system complexity, convergence time challenges, and need for interdisciplinary coordination
[13]	Conceptual framework integrating Artificial Immune Systems with Multi-Agent Systems and blockchain for industrial CPS security	Artificial Immune Systems, dendritic cell algorithms, negative selection methods, multi-agent systems, blockchain, PRISMA-based systematic literature review	Improved anomaly detection, data integrity, trust, context awareness, adaptive learning, and decentralized decision-making in industrial cyber–physical systems	Lack of scalability evaluation for large industrial networks and limited validation in real-world industrial scenarios

3. Proposed Framework Overview

The proposed framework focuses on developing a distributed multi agent reinforcement learning architecture capable of supporting autonomous decision making in dynamic environments. The framework consists of multiple intelligent agents that interact with a shared environment and learn optimal policies through continuous observation and reward driven learning.

The architecture is composed of several interconnected components including the environment layer, agent interaction layer, learning module, and decision execution module. The environment layer represents the dynamic system where agents operate and continuously observe state information. Each agent collects environmental observations and processes them through a reinforcement learning model to determine the most appropriate action.

The learning module enables agents to adapt their decision strategies by evaluating the reward obtained from previous actions. Through repeated interaction with the environment, agents gradually improve their policies and coordinate with neighboring agents to achieve better system level performance.

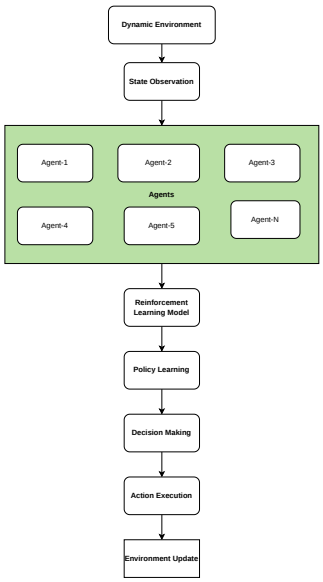

Figure 1: Overview of the proposed multi agent reinforcement learning framework for distributed autonomous decision making

4. System Model

The proposed framework considers a distributed environment consisting of multiple intelligent agents that interact with a dynamic environment to learn optimal decision policies. Each agent observes the environmental state and selects an action based on a reinforcement learning policy that evolves over time through reward driven learning. The environment continuously changes due to agent actions and external factors, which requires agents to adapt their strategies in order to maintain efficient decision making. The system model therefore represents the interaction among agents, environmental states, actions, and reward signals that guide the learning process.

Let the global environment be represented by a state space that captures the relevant features of the dynamic system. The state observed at time step t is defined as

$$S_t = \{s_1, s_2, s_3, \dots, s_n\} \tag{4.1}$$

where S_t represents the complete environmental state that can be observed by the agents. The formulation in Equation 4.1 allows the system to represent complex environmental conditions using multiple state variables.

Each agent selects an action from an available action set based on its current policy. The selected action at time step t is expressed as

$$a_t = \pi(S_t) \tag{4.2}$$

where π represents the decision policy that maps states to actions. The action defined in Equation 4.2 determines how an agent interacts with the environment.

After executing an action, the agent receives a reward signal that evaluates the effectiveness of the decision. The reward function is defined as

$$R_t = f(S_t, a_t) \tag{4.3}$$

The reward formulation in Equation 4.3 encourages agents to learn policies that maximize long term performance.

The cumulative return obtained by an agent over multiple time steps is defined as

$$G_t = \sum_{k=0}^{\infty} \gamma^k r_{t+k} \tag{4.4}$$

where γ represents the discount factor that determines the importance of future rewards. The return defined in Equation 4.4 enables the learning algorithm to evaluate long term decision outcomes.

To estimate the expected reward of state action pairs, a value function is introduced as

$$Q(S_t, a_t) \tag{4.5}$$

The value function defined in Equation 4.5 estimates the expected cumulative reward for executing an action in a given state.

The learning process updates the value function using a temporal difference update rule expressed as

$$Q(S_t, a_t) = Q(S_t, a_t) + \alpha[r_t + \gamma \max Q(S_{t+1}, a) - Q(S_t, a_t)] \tag{4.6}$$

The update rule shown in Equation 4.6 enables agents to iteratively improve their decision policies based on experience.

In multi agent environments, joint actions of multiple agents must be considered. The joint action set is defined as

$$A_t = \{a_1, a_2, \dots, a_n\} \tag{4.7}$$

The joint action representation in Equation 4.7 allows coordination among agents in distributed environments.

The state transition of the environment after an action is represented as

$$S_{t+1} = T(S_t, A_t) \tag{4.8}$$

The transition model defined in Equation 4.8 describes how the environment evolves over time.

The objective of the learning process is to maximize the expected reward defined as

$$J(\theta) = E[R] \tag{4.9}$$

The optimization objective defined in Equation 4.9 guides the reinforcement learning process toward improved decision policies.

5. Phase 1 Environment Modeling

The first phase focuses on modeling the dynamic environment in which multiple intelligent agents operate. The environment represents a complex system where environmental states change continuously due to agent actions and external influences. Each agent observes the environment through a state observation mechanism that captures relevant features required for decision making. The objective of this phase is to construct a structured representation of the environment that allows agents to interact effectively and collect meaningful observations for the learning process.

Algorithm 1 Environment Initialization and State Observation

Require: Number of agents N, state dimension d, observation set $\mathcal{O} = \{o_1, o_2, \dots, o_n\}$

Ensure: Normalized state vector $\hat{S}_t$ transmitted to all agents

Initialize dynamic environment parameters $\Phi \leftarrow \{\phi_1, \phi_2, \dots, \phi_k\}$
Define global state space $\mathcal{S} \subseteq \mathbb{R}^d$; set time step $t \leftarrow 0$
Initialize state vector $S_t = \{s_1, s_2, \dots, s_n\} \in \mathcal{S}$ (Eq. 4.1)
Assign agent i to observation location ℓ_i for all $i \in \{1, \dots, N\}$
Collect raw observations: $o_i \leftarrow \textsc{Observe}(\ell_i, \Phi)$ for each agent i
Aggregate observations: $S_t \leftarrow \frac{1}{N}\sum_{i=1}^{N} o_i$
Normalize state variables: $\hat{s}_j \leftarrow \dfrac{s_j - \mu_j}{\sigma_j}$ for all $j \in \{1, \dots, d\}$
Store processed state: $\mathcal{M} \leftarrow \mathcal{M} \cup \{\hat{S}_t\}$
Broadcast $\hat{S}_t$ to all N agents via shared memory buffer $\mathcal{B}$
Evaluate stability: $\delta_t \leftarrow \|S_t - S_{t-1}\|_2$; flag if $\delta_t > \epsilon_{\text{thresh}}$
Update environment parameters: $\Phi \leftarrow \Phi + \eta \cdot \nabla_\Phi \mathcal{L}(\hat{S}_t)$
Verify consistency: **assert** $\hat{S}_t \in [-1,\ 1]^d$
Compute feature vector: $\mathbf{f}_t \leftarrow W_{\text{enc}}\,\hat{S}_t + \mathbf{b}$, $W_{\text{enc}} \in \mathbb{R}^{d\times d}$
Validate feature map: **if** $\|\mathbf{f}_t\|_2 < \tau$ **then** re-initialize $S_t \leftarrow S_0$
Construct full state vector: $\mathbf{v}_t \leftarrow [\hat{S}_t \,\|\, \mathbf{f}_t] \in \mathbb{R}^{2d}$
Transfer $\mathbf{v}_t$ to the learning module $\mathcal{L}_{\text{MARL}}$
Monitor environment change: $\Delta_t \leftarrow S_{t+1} - S_t$; store in $\mathcal{M}$
Update observation matrix: $\Omega_t \leftarrow \Omega_{t-1} + \Delta_t \Delta_t^{\top}$
Cache state representation: $\mathcal{M}[t] \leftarrow (\mathbf{v}_t, \Phi, \delta_t)$ for future iterations
return Normalized state vector $\mathbf{v}_t$

6. Phase 2 Multi Agent Learning

The second phase focuses on enabling agents to learn optimal policies through reinforcement learning. Each agent interacts with the environment by observing system states and selecting actions according to a learning policy. The learning mechanism continuously improves decision strategies by evaluating reward signals obtained after executing actions. Through repeated interaction with the environment, agents gradually adapt their policies to maximize cumulative rewards. The multi agent learning process therefore allows the system to discover cooperative strategies that improve overall system efficiency and stability in dynamic environments.

Algorithm 2 Multi-Agent Reinforcement Learning Training

Require: Policy parameters θ_i for each agent $i \in \{1, \dots, N\}$, learning rate α, discount factor $\gamma \in [0, 1)$, exploration rate $\varepsilon \leftarrow 1.0$, replay buffer $\mathcal{D}$

Ensure: Optimized joint policy $\pi^* = \{\pi_1^*, \dots, \pi_N^*\}$

Initialize $Q_i(s, a; \theta_i) \leftarrow 0$, $V_i(s) \leftarrow 0$

Set $\varepsilon_{\min} \leftarrow 0.01$, decay $\lambda \leftarrow 0.995$

Observe initial state $S_0 \in \mathcal{S}$; set $t \leftarrow 0$

for each episode $e = 1, 2, \dots, E$ **do**

Observe state: $S_t^i \leftarrow \textsc{Observe}(\mathcal{E}, i)$

Select action using ε-greedy:

$$a_t^i = \begin{cases} \arg\max_a Q_i(S_t^i, a; \theta_i), & \text{w.p. } 1 - \varepsilon \\ \text{Uniform}(\mathcal{A}), & \text{w.p. } \varepsilon \end{cases}$$

Form joint action: $A_t = \{a_t^1, \dots, a_t^N\}$

Execute A_t, observe $S_{t+1} = T(S_t, A_t)$

Compute reward: $R_t^i = f(S_t, A_t)$

Store transition:
$\mathcal{D} \leftarrow \mathcal{D} \cup \{(S_t, A_t, R_t, S_{t+1})\}$

Sample mini-batch $\mathcal{B} \sim \mathcal{D}$

Compute TD target:

$$y_t = R_t^i + \gamma \max_{a'} Q_i(S_{t+1}, a'; \theta_i)$$

Update Q-function:

$$Q_i(S_t, a_t^i) \leftarrow Q_i(S_t, a_t^i) + \alpha \left[R_t^i + \gamma \max_{a'} Q_i(S_{t+1}, a') - Q_i(S_t, a_t^i) \right]$$

Update policy parameters:

$$\theta_i \leftarrow \theta_i + \alpha \nabla_{\theta_i} J(\theta_i)$$

where

$$J(\theta_i) = \mathbb{E}\left[\sum_{k=0}^{\infty} \gamma^k r_{t+k} \right]$$

Evaluate improvement: $\Delta J \leftarrow J(\theta_i^{new}) - J(\theta_i^{old})$

Anneal exploration: $\varepsilon \leftarrow \max(\varepsilon_{\min}, \varepsilon \cdot \lambda)$

Synchronize: $\bar{\theta} \leftarrow \frac{1}{N} \sum_{i=1}^{N} \theta_i$

Check convergence: **if** $\|Q_t - Q_{t-1}\|_2 < \delta_{\text{conv}}$ **then break**

Log $(\theta_i, J(\theta_i), \varepsilon)$

Compute return:

$$G_t = \sum_{k=0}^{T-t} \gamma^k R_{t+k}^i$$

end for

return $\{\theta_1^*, \dots, \theta_N^*\}$

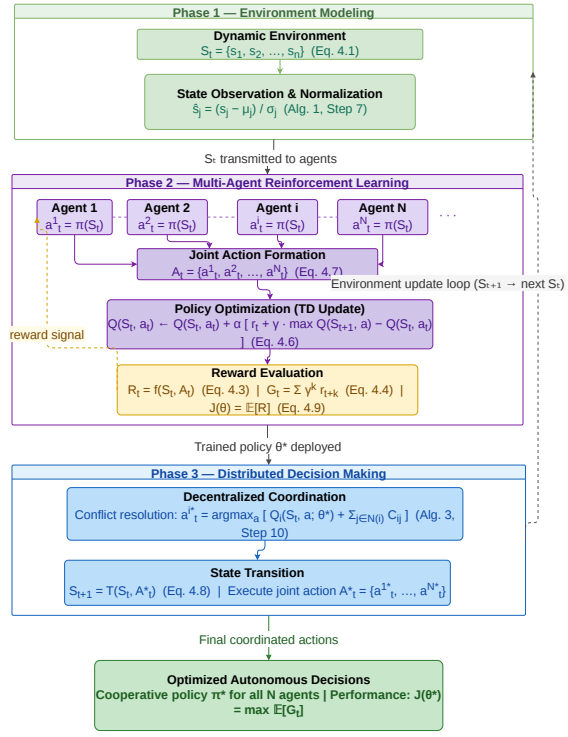

Figure 2: Overview of the proposed multi agent reinforcement learning framework for distributed autonomous decision making

The learning architecture shown in Fig. **??** illustrates the interaction between agents, environment states, and policy update mechanisms used to improve decision performance.

7. Phase 3 Distributed Decision Making

The final phase focuses on distributed decision making where agents independently determine their actions while considering the influence of other agents operating in the environment. This decentralized coordination mechanism improves system scalability and allows the framework to operate efficiently without relying on centralized control. Each agent evaluates its learned policy together with local environmental observations to select an action that maximizes expected reward while maintaining system stability.

Algorithm 3 Distributed Decision Coordination

Require: Trained policy parameters θ_i^* for all $i \in \{1, \dots, N\}$, coordination threshold ρ, neighborhood set $\mathcal{N}(i)$

Ensure: Coordinated joint action A_t^* maximizing $J(\theta) = \mathbb{E}[R]$ (Eq. 4.9)

1: Load trained parameters θ_i^* for each agent i; set $t \leftarrow 0$
2: Observe current state: $S_t = \{s_1, s_2, \dots, s_n\} \in \mathcal{S}$ (Eq. 4.1)
3: Collect local observation for each agent: $o_t^i \leftarrow \textsc{Observe}(\mathcal{E}, i)$
4: Evaluate policy output: $\pi_i^*(S_t) \leftarrow \arg\max_{a \in \mathcal{A}} Q_i(S_t, a; \theta_i^*)$ (Eq. 4.2)
5: Generate candidate actions: $\tilde{a}_t^i \leftarrow \pi_i^*(S_t)$ for all $i \in \{1, \dots, N\}$
6: Estimate expected reward: $\hat{R}_t^i \leftarrow Q_i(S_t, \tilde{a}_t^i; \theta_i^*)$ (Eq. 4.5)
7: Exchange coordination messages: $m^i \leftarrow (\tilde{a}_t^i, \hat{R}_t^i)$ to $j \in \mathcal{N}(i)$
8: Analyze interaction: $C_{ij} \leftarrow \nabla_{a^j} Q_i(S_t, A_t; \theta_i^*)$ for $j \in \mathcal{N}(i)$
9: Resolve conflicts: **if** $\|\tilde{a}_t^i - \tilde{a}_t^j\|_2 > \rho$ **then** $\tilde{a}_t^i \leftarrow \tilde{a}_t^i - \beta C_{ij}$
10: Select optimal action: $a_t^{i*} \leftarrow \arg\max_a \left[Q_i(S_t, a; \theta_i^*) + \sum_{j \in \mathcal{N}(i)} C_{ij}\right]$
11: Form joint action: $A_t^* = \{a_t^{1*}, a_t^{2*}, \dots, a_t^{N*}\}$ (Eq. 4.7)
12: Execute A_t^*; transition to $S_{t+1} = T(S_t, A_t^*)$ (Eq. 4.8)
13: Monitor system response: $\Delta R_t \leftarrow R_t - \mathbb{E}[R_{t-1}]$; log if $|\Delta R_t| > \xi$
14: Evaluate performance: $\mathcal{P}_t \leftarrow \frac{1}{N} \sum_{i=1}^{N} Q_i(S_t, a_t^{i*}; \theta_i^*)$
15: Adapt coordination: **if** $\mathcal{P}_t < \mathcal{P}_{t-1}$ **then** $\rho \leftarrow \rho \cdot (1 - \mu)$
16: Update policy: **if** $\|S_{t+1} - S_t\|_2 > \epsilon$ **then** $\theta_i^* \leftarrow \theta_i^* + \alpha \nabla_{\theta_i} J(\theta_i)$
17: Record outcomes: $\mathcal{D}_{\log} \leftarrow \mathcal{D}_{\log} \cup \{(S_t, A_t^*, R_t, \mathcal{P}_t)\}$
18: Increment $t \leftarrow t + 1$; repeat from line 2 for next state S_{t+1}
19: Continue monitoring: **while** $t < T_{\max}$ **and** not converged **do** repeat
20: **return** Final coordinated decision output A_t^* and performance log $\mathcal{D}_{\log}$

8. Results and Evaluation

8.1. Performance Analysis

The classification performance of the proposed framework is evaluated using accuracy as the primary metric. The experimental results demonstrate that the proposed multi agent reinforcement learning framework achieves superior classification accuracy compared to the baseline approaches [28], [10], [35], [6], [19] and [11]. The improved accuracy indicates that the agents are capable of making better coordinated decisions in dynamic environments. The comparative accuracy performance of the proposed framework and the existing approaches is illustrated in Fig. 3.

Precision evaluates the correctness of positive predictions produced by the agents during decision making. Higher precision indicates fewer incorrect decisions and improved reliability of the learning framework. The proposed framework achieves better precision compared with the baseline models which demonstrates improved policy learning. The comparative precision performance of all methods is shown in Fig. 4.

Recall measures the ability of the framework to correctly identify relevant events within the environment. Higher recall indicates that the learning agents successfully capture important environmental states. The proposed model demonstrates improved recall performance compared with baseline approaches which confirms effective learning behavior. The recall comparison of different models is illustrated in Fig. 5.

The F1 score represents the balance between precision and recall and is widely used to evaluate classification performance. A higher F1 score indicates improved reliability of the decision making system. The proposed framework achieves the highest F1 score among all evaluated approaches which confirms balanced performance. The comparison results are presented in Fig. 6.

Decision latency measures the time required for the system to generate an action after observing an environmental state. Lower latency is important for real time autonomous systems. The proposed framework demonstrates reduced decision latency compared with baseline approaches which improves responsiveness. The latency comparison results are presented in Fig. 7.

Training convergence analysis evaluates how quickly the learning algorithm reaches stable performance during training. Faster convergence indicates better sample efficiency and improved learning capability. The proposed framework converges earlier compared with baseline models which indicates efficient reinforcement learning behavior. The convergence comparison is illustrated in Fig. 8.

The reward progression during training reflects the learning stability of the reinforcement learning model. As the number of episodes increases the reward gradually improves until reaching a stable level which indicates successful policy optimization. The reward progression of the proposed framework during training is presented in Fig. 9.

Training loss represents the difference between predicted and optimal actions during the learning process. A decreasing loss value indicates successful policy learning and improved optimization. The loss gradually decreases as training progresses which demonstrates stable learning behavior of the proposed framework. The loss trend is shown in Fig. 10.

The scalability of the framework is evaluated by increasing the number of agents participating in the system. As the number of agents increases the coordination complexity also increases which may affect performance. The success rate of the system under different agent counts is illustrated in Fig. 11. The results show that the framework maintains acceptable performance even with larger agent populations.

Throughput measures the number of tasks successfully processed by the system within a given time period. Higher throughput indicates improved system efficiency.

The throughput performance under different agent counts is presented in Fig. 12. The results demonstrate that the system maintains efficient task processing under distributed conditions.

The effect of increasing agents on decision latency is further evaluated to understand system scalability. As the number of agents increases communication overhead may affect decision response time. The latency variation with agent count is illustrated in Fig. 13. The proposed framework demonstrates acceptable latency growth with increasing agents.

Resource utilization analysis evaluates how effectively computational resources are used as the number of agents increases. Efficient resource usage is essential for distributed autonomous systems. The utilization trend under varying agent counts is illustrated in Fig. 14. The results demonstrate reasonable resource efficiency of the proposed framework.

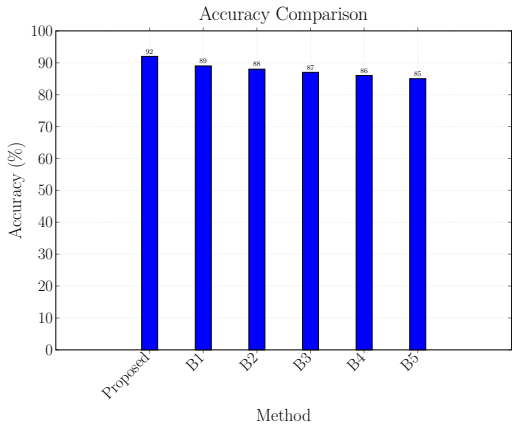

Figure 3: Comparison of classification accuracy of the proposed framework with baseline methods [28], [10], [35], [6], [19] and [11]. The proposed framework achieves higher accuracy which indicates improved decision making capability in distributed environments.

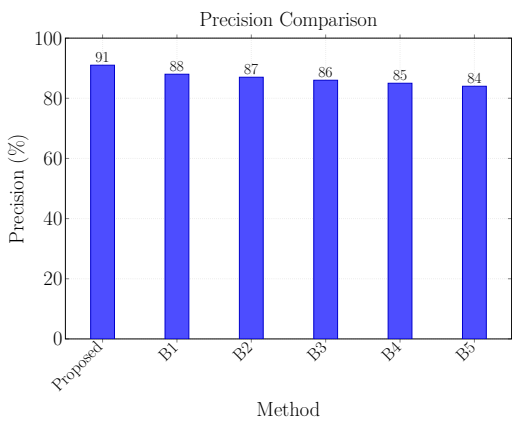

Figure 4: Comparison of precision of the proposed framework with baseline approaches [28], [10], [35], [6], [19] and [11]. Higher precision demonstrates improved correctness of agent level decisions.

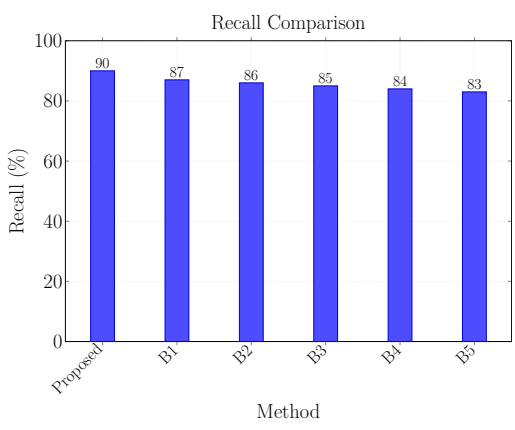

Figure 5: Recall comparison between the proposed framework and baseline methods [28], [10], [35], [6], [19] and [11] showing improved event detection capability.

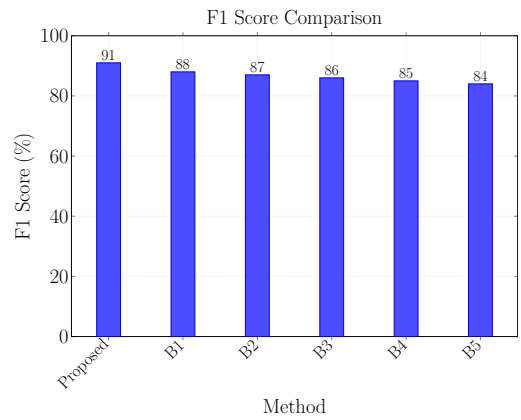

Figure 6: Comparison of F1 score between the proposed framework and baseline models [28], [10], [35], [6], [19] and [11] demonstrating balanced decision performance.

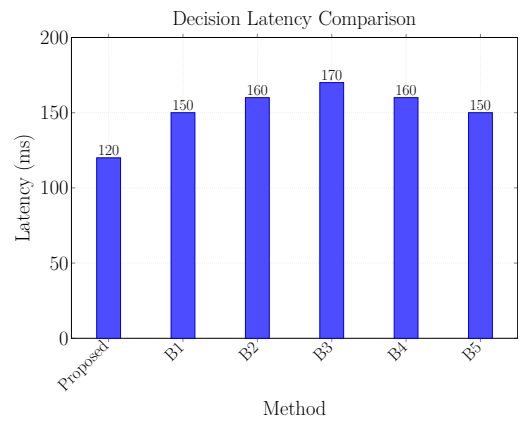

Figure 7: Comparison of decision latency of the proposed framework with baseline models [28], [10], [35], [6], [19] and [11] showing faster decision response.

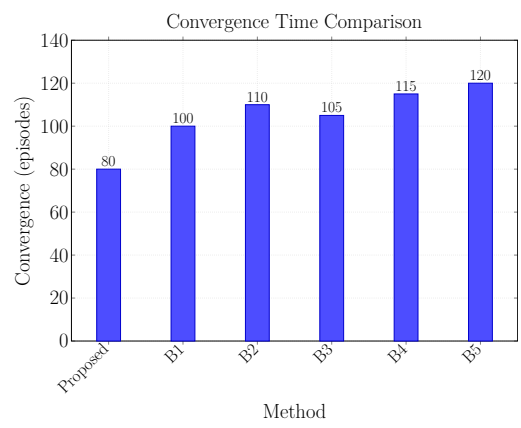

Figure 8: Training convergence comparison of the proposed framework with baseline methods showing faster learning behavior.

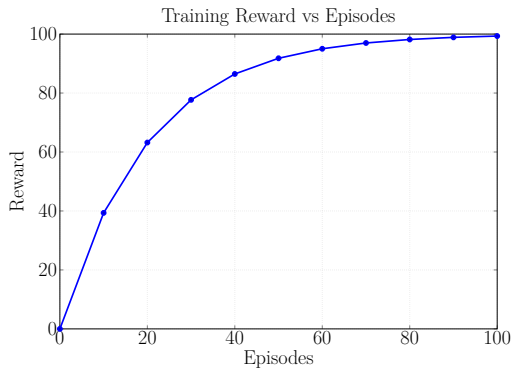

Figure 9: Reward progression during training showing improved learning stability of the proposed framework across training episodes.

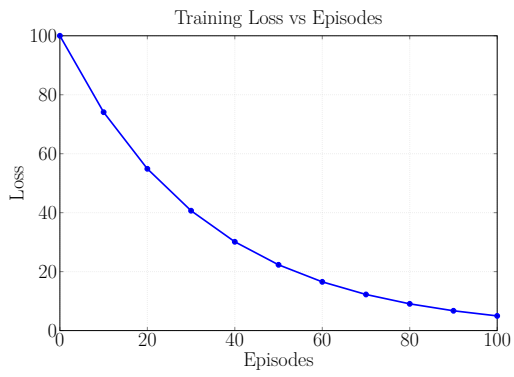

Figure 10: Training loss reduction across episodes showing stable optimization behavior of the proposed reinforcement learning model.

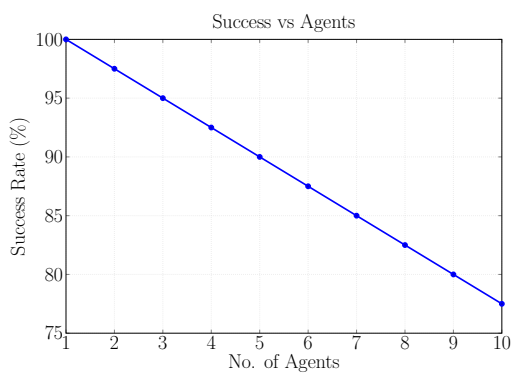

Figure 11: Success rate analysis of the proposed framework with increasing number of agents demonstrating scalability behavior.

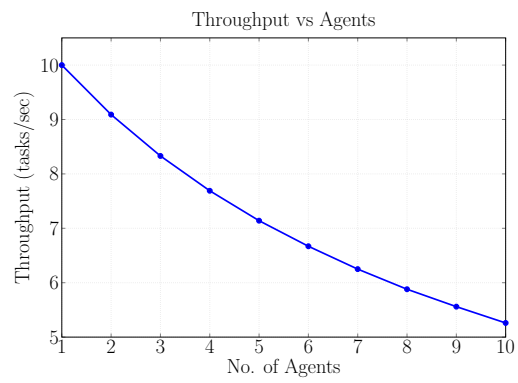

Figure 12: Throughput comparison of the proposed framework under varying number of agents showing system processing capability.

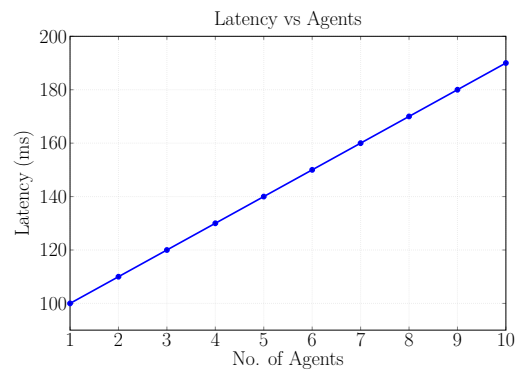

Figure 13: Variation of decision latency with increasing number of agents showing scalability impact on response time.

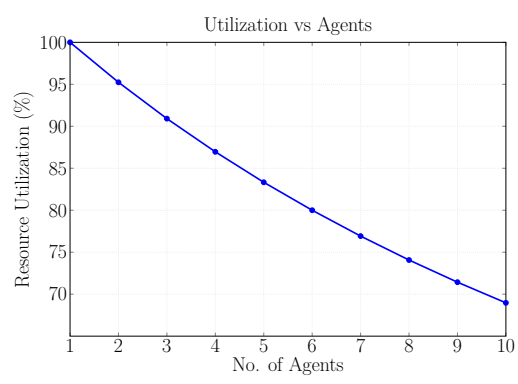

Figure 14: Resource utilization analysis of the proposed framework with varying number of agents demonstrating system efficiency.

9. Conclusion and Future Work

This study presented an overview of distributed autonomous decision making systems and highlighted the importance of multi agent reinforcement learning for enabling intelligent coordination among autonomous agents operating in dynamic environments. The literature analysis demonstrated that reinforcement learning based frameworks can significantly improve adaptability, collaboration, and decision efficiency in complex systems where centralized control is not feasible. However, existing approaches still face several challenges such as scalability limitations, communication overhead, unstable training behavior, and difficulties in handling rapidly changing environments. These limitations indicate the need for more efficient coordination mechanisms and adaptive learning strategies that can support large scale multi agent interactions. Future work will focus on the development of a robust multi agent reinforcement learning framework that integrates efficient policy learning, improved agent communication strategies, and scalable distributed training mechanisms. In addition, experimental validation using simulated dynamic environments will be conducted to evaluate the performance of the proposed framework in terms of learning efficiency, convergence behavior, and decision accuracy.

Declarations

Ethical Approval

This manuscript reports studies that do not involve human participants, human data, human tissue, or animals.

Conflict of Interest

The authors have no conflict of interest to declare that are relevant to the content of this article.

Authors' Contributions

S. Gopikrishnan contributed to the conceptualization, Formal analysis, drafting the original manuscript, and designing the experimental protocols. Author-2 was responsible for Conceptualization, methodology, software implementation, and dataset curation. Author3 contributed to conceptualization, supervision, and in-depth analysis of the experimental results. Author-3 handled the review and editing of the final draft, as well as the overall validation of the study results.

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Availability of data and materials

Data sharing is not applicable to this article, as no new data were created or analyzed in this study.

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